Problem 9 – GPT-6 Astra

Abstract

We prove the asserted constancy using rigidity of Teichmüller curves and the boundary structure theorem for totally geodesic subvarieties. An important preliminary observation is that the compatibility hypothesis forces every positive-dimensional boundary locus in a stratum with at least two nodes to remain fixed set-theoretically. This observation also handles the possibility that a cusp of one boundary curve lies on another positive-dimensional boundary locus.

1 Conventions and geometric input

We use the usual moduli-theoretic conventions: subvarieties are algebraic, while their dependence on the disc parameter may be holomorphic rather than algebraic. Smooth families and local arguments on moduli spaces are understood orbifold-locally. Thus auxiliary level structures and local labels for components and branches of nodes may be introduced. The conclusion is a statement on the original moduli space and is unaffected by these auxiliary choices.

Here is one consequence of the smooth-family assumption that will be used explicitly. Suppose that x is an isolated point of

$$Z_0(G) := \overline{V_0} \cap \Delta_G.$$

In a local marked chart at x, the family

$$\{(z,t) : z \in Z_t(G)\}, \qquad Z_t(G) := \overline{V_t} \cap \Delta_G,$$

has relative dimension zero and is a submersion to the parameter disc. The holomorphic inverse function theorem therefore gives a unique local section through the chosen marked point x. In particular, two sections through that marked point coincide after shrinking the parameter disc. Introducing such charts is legitimate for hypothesis (i), which is a smooth-family hypothesis. We will *not* assume that hypothesis (ii) holds for individual branches in a refined or marked stratification; it will be used only in the original moduli space.

We recall the following consequences of the boundary structure theorem. A component of the normalization of a nodal surface will be called *active* for a boundary locus if the projection of that locus to the corresponding pointed moduli space is nonconstant.

Theorem 1.1 (Boundary structure facts)**.** *Let $N \subset \mathcal{T}_{g,n}$ be an algebraic totally geodesic submanifold. The following statements hold.*

1. *Its intersection with a stratum of augmented Teichmüller space is semisimple. In particular, for a one-dimensional boundary locus, every nonconstant factor projection is a Teichmüller disc upstairs and is a local isometry on the corresponding normalized orbifold curve downstairs.*

2. *Pinchable curves have an N-equivalence relation. A pinched multicurve is a union of entire equivalence classes. If it contains r classes, the corresponding boundary locus has dimension at most $\dim N - r$. Any one class can be unpinched while all the other classes remain pinched.*

3. *For a positive-dimensional simple boundary factor, the union of the boundary curves of its active subsurfaces is a union of entire N-equivalence classes.*

References. These are, respectively, Theorem 1.3; Theorem 1.6(4), Corollary 3.17 and the dimension estimate in the proof of Theorem 3.9; and Lemma 6.1(3) of Arana–Herrera–Wright [1]. Their numbering here refers to version 2 of that paper. For the local-isometry assertion in the first statement, see also Benirschke–Dozier–Rached [2, Theorem 1.7 and Proposition 9.2]. □

We need the following useful consequence of the third statement.

Lemma 1.2 (Every node is seen by an active component)**.** *Let $W \subset \mathcal{M}_g$ be a totally geodesic surface, and let E be a one-dimensional irreducible component of its intersection with a boundary stratum. Then every node of the corresponding stable curves is incident to at least one component that is active for E.*

Proof. Choose a lift $N \subset \mathcal{T}_g$ of W, and a boundary locus upstairs covering E. If its pinched multicurve has r equivalence classes, Theorem 1.1(2) gives

$$1 \leq 2 - r.$$

Since this is a boundary locus, $r \geq 1$, so $r = 1$.

There is exactly one positive-dimensional simple factor, since the locus has dimension one. Its active subsurfaces have a nonempty union of boundary curves: the stable curve is connected, has a node, and the locus has at least one active component. By Theorem 1.1(3), this union consists of entire equivalence classes. There is only one such class, so the union is the whole pinched multicurve. This is precisely the assertion. □

2 Deformations of boundary curves

We will use rigidity of Teichmüller curves in its usual pointed and orbifold formulation.

Theorem 2.1 (Rigidity of Teichmüller curves)**.** *Let*

$$f_t : C_t \longrightarrow \mathcal{M}_{h,m}$$

be a holomorphic, smoothly trivial family of orbifold maps from finite-type curves. Suppose that f_0 is a local isometry for the complete finite-area hyperbolic metric on C_0 and the Teichmüller metric on the target. Then the family of pairs (C_t, f_t) is locally trivial.

Proof. For unpointed targets this is McMullen's rigidity theorem [8, Theorem 1]. We explain why the same argument applies to pointed targets.

Pass to finite covers to remove orbifold stabilizers, and identify monodromy using the smooth trivialization. For a nonperipheral conjugacy class $\gamma \in \pi_1(C_0)$, its monodromy is pseudo-Anosov, with translation length equal to the hyperbolic length $\ell_0(\gamma)$. Holomorphic maps contract the Teichmüller metric, by the Royden–Earle–Kra theorem [10, 6]. Consequently,

$$\ell_t(\gamma) \geq \ell_0(\gamma) \qquad \text{for every nonperipheral } \gamma.$$

The marked length-spectrum domination principle used in [8, proof of Theorem 1, fact (1)] implies that C_t and C_0 are conformally identical with their markings.

For completeness, the maps are then identical as marked maps. Lift them to equivariant maps from the same hyperbolic disc to $\mathcal{T}_{h,m}$. Equality in the preceding inequality along a hyperbolic

element's axis forces that axis to map isometrically onto the unique Teichmüller axis of its monodromy [3]. Holomorphic continuation identifies the lifted map with the original Teichmüller-disc parametrization, up to an automorphism of the source disc. Equivariance makes this automorphism commute with the source lattice. A non-elementary Fuchsian lattice has trivial centralizer in $\mathrm{PSL}_2(\mathbb{R})$: commuting with two hyperbolic elements with different axes fixes at least three boundary points. The automorphism is therefore the identity.

Finally, the universal property of marked Teichmüller space makes these identifications holomorphic in the parameter. Thus the family of pairs is locally trivial. □

Lemma 2.2 (Deforming a curve in a product)**.** *Suppose that E_t is a holomorphic, smoothly trivial family of algebraic curves in a product of pointed moduli spaces. Suppose that every nonconstant factor projection of the normalized central curve is a Teichmüller local isometry. After passing to finite covers where necessary and shrinking the parameter disc, the following hold.*

1. *The normalized source curve can be identified with a fixed curve C, and every initially nonconstant factor map is independent of t.*
2. *Every initially constant factor map remains constant on C. Its value may depend holomorphically on t.*
3. *Each specified point of the compactification of C determines a holomorphic family of stable limits of E_t. Its dual graph is independent of t. Every pointed component belonging to the stable limit of an initially nonconstant factor map is fixed.*

Proof. At least one factor projection is nonconstant. Apply Theorem 2.1 to that projection to identify the source family with $C \times \Delta$. Apply the same theorem to the other initially nonconstant projections. The marked identifications of the source agree, proving the first assertion.

Consider an initially constant projection. Its orbifold monodromy has finite image. This image is unchanged under the deformation. After a finite cover of C killing that image, each deformed map lifts holomorphically to the target Teichmüller space. A Bers embedding realizes that space as a bounded domain [4]. The coordinates of the lift are bounded holomorphic functions on a compact Riemann surface minus finitely many points. They extend across the punctures and are constant. Hence the original factor map is constant. Its value depends holomorphically on t, as is seen by evaluating at a fixed point of C.

Let $\overline{C}$ be the smooth compactification of C. The algebraic factor maps extend to the corresponding Deligne–Mumford compactifications, with stable reduction or finite covers if needed. For a nonconstant factor the extended map is fixed; for a constant factor it is the moving smooth point just described. Composing their product with the clutching morphism gives the claimed family of stable limits. Clutching and the local stable-curve description are those of [5]. The nonconstant factors' stable limits are fixed, and the constant factors remain smooth pointed curves of the same topological type. Thus the dual graph does not change. □

Lemma 2.3 (A boundary curve has a further degeneration)**.** *A positive-dimensional algebraic Teichmüller curve is noncompact. Consequently, a one-dimensional boundary locus as in Theorem 1.1(1) has a stable limit in a strictly smaller boundary stratum.*

Proof. A nonzero finite-area quadratic differential generating a Teichmüller disc has a cylinder; see [7]. Rotate the differential so that this cylinder is horizontal, and apply Teichmüller flow in the direction that makes its modulus tend to infinity. Its core curve has extremal length tending to zero. By compactness of the thick part of moduli space [9], the resulting surfaces leave every compact subset of moduli space. An algebraic Teichmüller curve is therefore noncompact.

For a boundary curve in a product, apply this to a nonconstant factor projection. Compactify the normalized source and the factor maps. Some stable limit acquires an additional node, and hence lies in a strictly smaller stratum. □

3 A consequence of the compatibility hypothesis

The next observation uses the exact, set-theoretic formulation of hypothesis (ii).

Lemma 3.1 (Deep boundary curves cannot move)**.** *Let Δ_G be a boundary stratum whose curves have at least two nodes. For every t, every positive-dimensional irreducible component of $Z_t(G)$ is contained in $Z_0(G)$.*

Proof. Choose a strictly larger boundary stratum Δ_H whose closure contains Δ_G. Such a stratum exists: smooth one node while leaving at least one node unsmoothed.

Set

$$U = \overline{\mathcal{M}}_g \setminus \overline{V_0}.$$

For this open set, both sides of the equality in hypothesis (ii) are empty at $t = 0$. That hypothesis therefore implies, for every t,

$$Z_t(G) \cap U = \overline{Z_t(H)} \cap \Delta_G \cap U. \tag{1}$$

Since $\overline{V_t}$ is a surface, its boundary has dimension at most one. Thus $Z_t(H)$ has dimension at most one. Every one-dimensional irreducible component of $\overline{Z_t(H)}$ has a dense open subset in Δ_H; its intersection with the disjoint stratum Δ_G is consequently finite. Zero-dimensional components contribute no boundary points. It follows that

$$\overline{Z_t(H)} \cap \Delta_G$$

is finite.

Let E_t be a positive-dimensional irreducible component of $Z_t(G)$. If E_t were not contained in $\overline{V_0}$, then $E_t \cap U$ would be a nonempty open subset of an algebraic curve, and hence infinite. This contradicts (1). Therefore

$$E_t \subset \overline{V_0} \cap \Delta_G = Z_0(G).$$

□

Remark 3.2. The preceding proof is the only use of hypothesis (ii). In particular, no stronger assertion about preservation of individual incidences between branches is being substituted for the hypothesis in the question.

4 Proof of the assertion

Theorem 4.1. *Under the hypotheses of the question,*

$$\pi_i(\partial V_t) = \{Y_0\} \qquad \text{for every } t \in \Delta.$$

Proof. The function $t \mapsto Y_t$ is holomorphic. Indeed, locally choose a holomorphic section of the smooth family ∂V_t; composing with π_i gives Y_t, independently of the chosen section. It suffices to prove constancy near 0, since the identity principle then gives constancy on the connected disc.

Choose a one-dimensional component B_0 of ∂V_0, and let B_t be its continuation in the smoothly trivial family from hypothesis (i).

Step 1: Reduction to a one-node stratum.

If Γ has at least two edges, Lemma 3.1 gives

$$B_t \subset Z_0(\Gamma) = \partial V_0.$$

Since π_i is equal to Y_t on B_t and equal to Y_0 on ∂V_0, this already proves $Y_t = Y_0$.

We may therefore assume that Γ has one edge. It cannot be a nonseparating edge: that stratum has just one moduli-space factor, and constancy on that factor would contradict $\dim B_0 = 1$. Thus the node is separating. Write its vanishing curve as a. Its two pointed components will be denoted

$$Y_t \in \mathcal{M}_{h,1}, \qquad X \in \mathcal{M}_{g-h,1}.$$

The Y-projection of B_t is constant, whereas the X-projection of B_0 is nonconstant.

Theorem 1.1(1) and Lemma 2.2 identify the normalized source of B_t with a fixed curve and make its X-map independent of t.

Step 2: A cusp section.

By Lemma 2.3, choose a cusp of this fixed X-map. Let X^* be its stable pointed limit. It has at least one node. Lemma 2.2(3) gives a holomorphic section

$$p(t) \in Z_t(G')$$

obtained by attaching Y_t to the fixed pointed stable curve X^*. Here $\Delta_{G'}$ is a fixed stratum with at least two nodes. Put $p_0 = p(0)$.

Use local labels for the components and nodes of p_0. Along $p(t)$, all pointed components coming from X^* are fixed. Only the component Y_t can move.

The smooth locus $Z_0(G')$ has local dimension either zero or one at p_0. We treat these possibilities separately.

Step 3: A positive-dimensional boundary locus through p_0.

Suppose first that $Z_0(G')$ has local dimension one at p_0. Hypothesis (i) puts $p(t)$, for small t, on a positive-dimensional component of $Z_t(G')$. Lemma 3.1 then implies

$$p(t) \in Z_0(G').$$

Let E_0 be its local curve component through p_0.

Choose any node b of X^*. Both components incident to b, or the single component if b is a self-node, come from X^*; their pointed moduli are fixed along $p(t)$. On the other hand, Lemma 1.2 says that at least one of these components is active for E_0. Projection of E_0 to that factor is locally injective in a marked chart, by Theorem 1.1(1). Its value along $p(t)$ is fixed, so $p(t)$ itself is constant near 0. In particular, $Y_t = Y_0$.

Step 4: An isolated boundary point.

It remains to suppose that p_0 is isolated in $Z_0(G')$. Choose a marked lift of the degeneration to the closure of a lift $N \subset \mathcal{T}_g$ of V_0. Let $A = \{a\}$, and let B be the nonempty multicurve consisting of the additional nodes of X^*. Thus the nodes at p_0 correspond to

$$A \sqcup B.$$

Because A itself occurs as the complete pinched multicurve along B_0, it is an entire N-equivalence class. There can be at most two equivalence classes in $A \sqcup B$, by Theorem 1.1(2) and $\dim N = 2$. Consequently B is the other class.

Unpinch the class A, leaving B pinched. The unpinching assertion gives points arbitrarily close to p_0 in the stratum obtained by smoothing a. There is therefore a one-dimensional boundary component F_0 of V_0 in that stratum with

$$p_0 \in \overline{F_0}.$$

To justify the dimension assertion explicitly, the boundary of the surface has dimension at most one, and a zero-dimensional algebraic locus is finite and cannot supply points in that stratum arbitrarily approaching p_0.

Let Z denote the component obtained by smoothing a. It contains the original Y_0 and the component of X^* to which Y_0 was attached. The projection of F_0 to the moduli space of Z is nonconstant: its stable limit at the specified cusp has the node a, whereas a constant smooth pointed curve cannot have a nodal limit.

Let F_t be the continuation of F_0 supplied by hypothesis (i). If its ambient stratum is a finite quotient of a product, take its product cover and the corresponding finite cover of F_t. Apply Lemma 2.2. The Z-map is independent of t, and its stable limit at our specified cusp is independent of t. In particular, the pointed component inside that stable limit which is the original Y_0 remains exactly Y_0.

The same lemma supplies a cusp section

$$q(t) \in Z_t(G'), \qquad q(0) = p_0,$$

with the same local component and node labels. Since p_0 is isolated in the central stratum intersection, hypothesis (i) and the local inverse-function observation in Section 1 imply

$$q(t) = p(t)$$

near 0, in the chosen marked chart. Their specified Y-components agree. The Y-component of $q(t)$ is Y_0, while that of $p(t)$ is Y_t. Thus $Y_t = Y_0$ near 0.

This completes both cases. Descending any auxiliary markings or finite covers and applying the identity principle yields

$$\pi_i(\partial V_t) = \{Y_0\} \qquad (t \in \Delta).$$

□

Remark 4.2. Only the central surface and its central boundary curves were assumed to be totally geodesic. For $t \neq 0$, rigidity of the nonconstant projections of boundary curves follows from Theorem 2.1; total geodesicity of V_t is never used.

References

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